Lei Song
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Rutgers University
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EDUCATION

2023 **Ph.D. Geography**
Clark University, Worcester, MA, USA

2021 **M.A. Geography**
Clark University, Worcester, MA, USA

2015 **M.S. Applied Meteorology**
Nanjing University of Information Science & Technology, Nanjing, CHINA

2012 **B.S. Geographic Information Science**
Nanjing University of Information Science & Technology, Nanjing, CHINA

APPOINTMENTS

2025-Present **Assistant Professor,** Department of Geography, Rutgers University-New Brunswick, NJ, USA.

2023-2025 **Postdoctoral Research Fellow**, Department of Geography, University of California, Santa Barbara, CA, USA.

2023 **Postdoctoral Research Fellow**, School of Geographical Sciences and Urban Planning, Arizona State University, Tempe, AZ, USA.

GRANTS & FELLOWSHIPS

2026 - 2028 *Rutgers Research Council Award,* Rutgers University-New Brunswick, Reassessing thermal risk in a compound climatic context for global biodiversity
Amount: $15,000. (Role: **PI**)

2020 - 2023 *NASA Future Investigators in NASA Earth and Space Science and Technology (FINESST),* Combining Spatially-explicit Simulation of Animal Movement and Earth Observation to Reconcile Agriculture and Wildlife Conservation. Award No. 80NSSC20K1640.
Amount: $135,000. (Role: **FI**, PI: Lyndon Estes)

HONORS & AWARDS

2025 *Individualized Professional Skills Grant,* University of California, Santa Barbara
NASA-MSU Professional Enhancement Award, IALE-North America

2018 *The Pruser Dissertation Enhancement Awards*, Clark University
Harvard Design and Map Company Enhancement Award, Clark University

2015, 2009 *The First Prize Scholarship*, Nanjing University of Information Science & Technology

2011 *The Third Prize Scholarship*, Nanjing University of Information Science & Technology

PEER-REVIEWED ARTICLES

16. **Song, L.**, Frazier, A.E., Kedron, P., Araujo, D.S., Enquist, B.J., Maitner, B., Merow, C., Moulatlet, G.M., Nikolopoulos, E.I. and Roehrdanz, P.R. (2026). Explainable Artificial

Intelligence Reveals Spatially Divergent Effects of Global Change on Mammals. *Nature Communications*. https://doi.org/10.1038/s41467-026-78677-5

15. Frazier, A.E., Nelson, T., Kedron, P., Dodge, S., Murray, A., Vukomanovic, J., Franklin, J., Miller, J., Sarkar, D., Kinka, D., Brown, D.G., Devenish, A., Easterday, K., Gadsden, D., Herrero, H.V., Leiker, S., McManus, N., Medrano, A., Meerdink, S., Molitor, C., Papeş, M., Rathbone, V., Rodríguez-Buriticá, S., Roehrdanz, P.R., Rose, M.B., Singh, R., **Song, L.**, Syphard, A.D., Wyckoff, T.B., Youngs, Y., Zuther, S. (2026). Advancing conservation science and practice through a geographic approach: Platforms, personnel, and partnerships. *Conservation Science and Practice*, e70410. https://doi.org/10.1111/csp2.70410
14. **Song, L**, Frazier AE, Roehrdanz PR, Yang W, Krieger JM, McManus N, Urbano EA, Buritica SR, Willett D. Modeling climate adaptive habitat connectivity for conservation planning in Colombia. *Ecography*. 2026 Sep:e08789.
13. Frazier, A.E., McManus, N., **Song, L.**, Roehrdanz, P.R., Yang, W., Willett, D., Buriticá, S.R., Urbano, E.N. and Krieger, J.M., 2026. Scaling Spatial Planning Models to Support Conservation and Land Use Decisions under Climate Change. *Biodiversity Information Science and Standards*, 10, p.e182057.
12. Moulatlet, G.M., Boyle, B., Feng, X., Frazier, A., Hinojo-Hinojo, C., Maitner, B.S., Merow, C., Newman, E.A., Roehrdanz, P.R., **Song, L.**, Villalobos, F., Marquet, P.A., Svenning, J.-C. and Enquist, B.J. (2025). General laws of biodiversity: climatic niches predict plant range size and ecological dominance globally. *Proceedings of the National Academy of Sciences*, 122(46), e2517585122. www.pnas.org/doi/10.1073/pnas.2517585122
11. **Song, L.**, Frazier, A. E., Crawford, C. L., Estes, A. B., & Estes, L. D. (2025). Optimizing future cropland allocation in a biodiverse savanna by integrating agricultural benefits and ecological costs. *Landscape Ecology,* 40(11), 1–18.
10. **Song, L.**, Frazier, A. E., Estes, A. B., & Estes, L. D. (2025). A multi-scale approach for integrating species distribution models with landscape connectivity to identify critical linkage zones for African savanna elephants (*Loxodonta africana*). *Ecological Modelling*, 507, 111198. https://doi.org/10.1016/j.ecolmodel.2025.111198
9. Araujo, D. S., Enquist, B. J., Frazier, A. E., Merow, C., Roehrdanz, P. R., Moulatlet, G. M., Zvoleff, A., **Song, L.**, Maitner, B., & Nikolopoulos, E. I. (2025). Global Future Drought Layers Based on Downscaled CMIP6 Models and Multiple Socioeconomic Pathways. *Scientific Data*, 12(1), 295.
8. Khallaghi, S., Abedi, R., Abou Ali, H., Alemohammad, H., Dziedzorm Asipunu, M., Alatise, I., Ha, N., Luo, B., Mai, C., **Song, L.**, Wussah, A. O., Xiong, S., Yao, Y.-T., Zhang, Q., & Estes, L. D. (2025). Generalization Enhancement Strategies to Enable Cross-Year Cropland Mapping with Convolutional Neural Networks Trained Using Historical Samples. *Remote Sensing*, 17(3). https://doi.org/10.3390/rs17030474.
7. Frazier, A. E., & **Song, L.** (2025). Artificial Intelligence in Landscape Ecology: Recent Advances, Perspectives, and Opportunities. *Current Landscape Ecology Reports*, *10*(1), 1-13. https://doi.org/10.1007/s40823-024-00103-7
6. **Song, L.**, Estes, A. B., & Estes, L. D. (2023). A super-ensemble approach to map land cover types with high resolution over data-sparse African savanna landscapes. *International Journal of Applied Earth Observation and Geoinformation*, 116, 103152. https://doi.org/10.1016/j.jag.2022.103152
5. **Song, L.**, & Estes, L. (2023). ITSDM: Isolation forest-based presence-only species distribution modelling and explanation in R. *Methods in Ecology and Evolution*, 2041-210X.14067. https://doi.org/10.1111/2041-210X.14067

4. Estes, L. D., Ye, S., **Song, L.**, Luo, B., Eastman, J. R., Meng, Z., Zhang, Q., McRitchie, D., Debats, S. R., Muhando, J., Amukoa, A. H., Kaloo, B. W., Makuru, J., Mbatia, B. K., Muasa, I. M., Mucha, J., Mugami, A. M., Mugami, J. M., Muinde, F. W., … Caylor, K. K. (2022). High Resolution, Annual Maps of Field Boundaries for Smallholder-Dominated Croplands at National Scales. *Frontiers in Artificial Intelligence*, 4, 744863. https://doi.org/10.3389/frai.2021.744863
3. Elmes, A., Alemohammad, H., Avery, R., Caylor, K., Eastman, J., Fishgold, L., Friedl, M., Jain, M., Kohli, D., Laso Bayas, J., Lunga, D., McCarty, J., Pontius, R., Reinmann, A., Rogan, J., **Song, L.**, Stoynova, H., Ye, S., Yi, Z.-F., & Estes, L. (2020). Accounting for Training Data Error in Machine Learning Applied to Earth Observations. *Remote Sensing*, 12(6), 1034. https://doi.org/10.3390/rs12061034
2. Shi, Y., & **Song, L.** (2015). Spatial Downscaling of Monthly TRMM Precipitation Based on EVI and Other Geospatial Variables Over the Tibetan Plateau From 2001 to 2012. *Mountain Research and Development*, 35(2), 180–194. https://doi.org/10.1659/MRD-JOURNAL-D-14-00119.1
1. Shi, Y., **Song, L.**, Xia, Z., Lin, Y., Myneni, R., Choi, S., Wang, L., Ni, X., Lao, C., & Yang, F. (2015). Mapping Annual Precipitation across Mainland China in the Period 2001–2010 from TRMM3B43 Product Using Spatial Downscaling Approach. *Remote Sensing*, 7(5), 5849–5878. https://doi.org/10.3390/rs70505849

Manuscripts in review/revision

17. Yang, W., Frazier, A.E., Kedron, P., **Song, L.**, Wu, J. and Nelson, T. (Under review). Three-dimensional approaches to assessing terrestrial habitat for species conservation: a systematic review.
18. Yang, W., **Song, L.**, Roehrdanz, P.R., McManus, N., Willett, D., Krieger, J., Noguera-Urbano, E., Rodriguez-Buritica, S., Ramos, K., Blair, M., Cruz-Rodríguez, C., Muñoz-R, C. and Frazier, A.E. (In revision). A Geographically Replicable Framework for Area of Habitat (AOH) Mapping at National and Sub-National Scales.
19. Hinojo-Hinojo, C., **Song, L.**, Frazier, A.E., Nikolopoulos, E.I., Wainwright, H., Merow, C., Goulden, M., Falco, N., Araujo, D.S., Chacón-Labella, J., Bohner, T., Chmurzynski, A., Hemingway, B., Maitner, B.S., Moulatlet, G.M., Wang, J. and Enquist, B.J. (In revision). Remotely sensed evidence of long-term global shifts in leaf mass per area.
20. Bohner, T., Duncanson, L., Nikolopoulos, E., Frazier, A., Araujo, D., Brock, C., Cui, D., Hinojo-Hinojo, C., Krieger, J.M., Maitner, B.S., Moulatlet, G.M., Roehrdanz, P.R., **Song, L.**, Merow, C. and Enquist, B.J. (In revision). Forecasting global drought exposure reveals biome-specific risk to Earth's forests and tallest trees.
21. Moulatlet, G.M., Capparelli, M., Thomas, C., Boyle, B., Feng, X., Frazier, A., Hinojo-Hinojo, C., Herrera-Perez, J., Kajiki, L., Lechner, A., Maitner, B.S., Newman, E.A., Nikolopoulos, E., Roehrdanz, P.R., **Song, L.**, Valencia-Rodriguez, D., Yang, W., Merow, C., Silman, M., Villalobos, F., Macklin, M. and Enquist, B.J. (In revision). Mining in the Amazon Basin causes metal contamination of freshwater systems and threatens biodiversity. *Global Change Biology*. Preprint: https://doi.org/10.1101/2025.05.21.654597

Manuscript in prep

22. Montes-Bojorquez, M.G., Robles-Zazueta, C.A., Tinoco-Ojanguren, C., Enquist, B.J., Frazier, A.E., Maitner, B.S., Massaine Moulatlet, G., Romo-Leon, J.R., **Song, L.** and Hinojo-Hinojo, C. (To submit). Challenges and opportunities in detecting leaf water and carotenoid content across biomes from satellite multispectral indices. Preprint: https://doi.org/10.1101/2025.11.01.686024

PEER-REVIEWED CONFERENCE PROCEEDINGS

2024 **Song, L.**, Frazier, A.E., Kedron, P., Araujo, D.S., Cui, D., Enquist, B.J., Maitner, B., Merow, C., Moulatlet, G.M., Nikolopoulos, E.I. and Roehrdanz, P.R., (2024). Explainable artificial intelligence to interpret spatially-explicit impacts of future climate change on species distribution. In *I-GUIDE Forum*. October 2024.

TEACHING EXPERIENCE

Instructor

01:450:214 - Conservation, Rutgers University-New Brunswick	2026 Fall
01:450:101 - Earth Systems, Rutgers University-New Brunswick	2025 Fall & Spring, 2026 Fall
01:450:320 - Spatial Data Analysis, Rutgers University-New Brunswick	2026 Spring
GEOG 274 - Conservation GIS, University of California, Santa Barbara (Co-instructor)	2025 Spring
GEOG 246/346 - Geospatial Analysis with R, Clark University	2021 Spring

Teaching Assistant

GEOG 246/346 - Geospatial Analysis with R, Clark University	2020 Spring & Fall
GEOG 293/383 - Introduction to Remote Sensing, Clark University	2017 Fall
Principles of Remote Sensing, Nanjing University of Information Science & Technology	2015 Fall
IDL programming, Nanjing University of Information Science & Technology	2014 Fall

PROFESSIONAL EXPERIENCE

2018 - 2020 Graduate Research Assistant, Clark University, Worcester, USA

- *Mapping Africa project*

2012 - 2015 Graduate Research Assistant, Nanjing University of Information Science & Technology, Nanjing, CHINA

INVITED TALKS AND PRESENTATIONS

2026 **Song, L.**, Uncovering Pathways, Risks, and Opportunities for Species in a Changing World. *Invited seminar talk for Ecology & Evolution Graduate program Seminar, Rutgers University-New Brunswick, NJ*. Mar 2026.

2025 **Song, L.**, Protecting biodiversity under global change. *Invited seminar talk for graduate course Advanced GIScience Seminar, University of New Mexico, Albuquerque, NM*. Sept 2025.

2023 **Song, L.**, Combining Earth Observation and Animal Movement to Reconcile Agriculture and Wildlife Conservation. *Center for Spatial Studies and Data Science, University of California, Santa Barbara*. 01 Nov. 2023

Song, L., A super-ensemble approach to map land cover types with high resolution over data-sparse African savanna landscapes. *8th NICFI Satellite Data Program Scientific Research Roundtable*. Online. 01 June 2023

CONFERENCE PRESENTATIONS & POSTERS

2026 **Song, L.** How precipitation modulates thermal effects across vertebrates. In *UCGIS Symposium, June 2026*.

Song, L., Frazier, A.E., Kedron, P., Araujo, D.S.A., Cui, D., Enquist, B.J., Hinojo-Hinojo, C., Krieger, J.M., Maitner, B.S., Merow, C., Moulatlet, G.M., Nikolopoulos, E.I. and Roehrdanz, P.R. (2026). Conservation planning with explainable species distribution modeling. In *IALE-North America Annual Meeting, April 2026*.

Song, L., Frazier, A.E., Kedron, P., Araujo, D.S.A., Cui, D., Enquist, B.J., Hinojo-Hinojo, C., Krieger, J.M., Maitner, B.S., Merow, C., Moulatlet, G.M., Nikolopoulos, E.I. and Roehrdanz, P.R. (2026). Spatially divergent impacts of future global change on mammals. In *AAG Annual Meeting, March 2026*.

2025 **Song, L.**, Kedron, P. (2025). Strategic Breakout: The Role of GIScience in Causal Inference. *UCGIS Symposium 2025, June 2025*.

Song, L., Frazier, A.E., Roehrdanz, P.R., Yang, W., Krieger, J., McManus, N., Noguera Urbano, E., Rodriguez Buriticá, S. and Willett, D. (2025). Modeling climate adaptive habitat connectivity for conservation planning. In *IALE-North America Annual Meeting*, *April 2025*.

2024 **Song, L.**, Frazier, A. E., Crawford, C. L., Estes, A. B., & Estes, L. D. (2024). Balancing agricultural benefits and ecological costs: pathways to address the land gap of agriculture in Tanzania. In *AAG Annual meeting, April 2024*.

2022 **Song, L.**, & Estes, L. D. (2022, December). Using a Nested Multi-Scale Method to Characterize Landscape Utilization and Conservation Status of African Savanna Elephant. In *AGU Fall Meeting Abstracts* (Vol. 2022, pp. B22D-1486).

Song, L., Estes, L., Estes, A., Using a nested multi-Scale method to characterize landscape utilization and conservation status of African Savanna Elephant, in *NASA Biological Diversity and Ecological Forecasting Team Meeting, September 20-22, 2022*.

Song, L., & Estes, L. *itsdm*: Isolation Forest-based presence-only species distribution modeling and explanation in R. In *2022 ESA Annual Meeting, August 14-19*.

Song, L., Estes, L., Luo, B., Estes, A., A super-ensemble approach to map land cover types with high resolution over data-sparse African savanna landscapes, in *the 4th International Electronic Conference on Remote Sensing, 25-27 January 2022*, online, **Best Poster Award**

2021 **Song, L.**, Estes, L., Luo, B., Estes, A., Land cover mapping in data-sparse regions, in *NASA Biological Diversity and Ecological Forecasting Team Meeting, October 19-21, 2021*, online

2020 **Song, L.**, Luo, B., Ye, S., Zhang, Q., & Estes, L. D. (2020, December). Using mixed labels and a multi-stage approach to map crop types over smallholder-dominated agricultural systems. In *AGU Fall Meeting Abstracts* (Vol. 2020, pp. GC034-04).

Song, L., & Estes, L. Broad-scale spatial distribution of African bush elephant (Loxodonta Africana) using combination of machine learning algorithms. In *2020 ESA Annual Meeting (August 3-6)*. ESA.

2019 Estes, L. D., Ye, S., **Song, L.**, Avery, R. B., McRitchie, D., Eastman, R., ... & Caylor, K. K. (2019, December). Improving maps of smallholder-dominated croplands through tight integration of human and machine intelligence. In *AGU Fall Meeting Abstracts* (Vol. 2019, pp. IN42A-04).

2018 Avery, R.B., Caylor, K.K., Estes, L.D., Eastman, R., Ye, S., **Song, L.**, Zhang, K., Xiong, S., McRitchie, D. and Woodard, T. (2018, December). A Convolutional Neural Network Approach to Segmenting Smallholder Agriculture. In *AGU Fall Meeting Abstracts* (Vol. 2018, pp. B31I-2597).

Estes, L.D., Debats, S.R., McRitchie, D., Eastman, R., **Song, L.**, Woodard, T., Xiong, S., Ye, S., Zhang, K., Avery, R.B. and Caylor, K.K. (2018, December). Using active learning to quantify how training data errors impact classification accuracy over smallholder-dominated agricultural systems. In *AGU Fall Meeting Abstracts* (Vol. 2018, pp. B22A-05).

TECHNOLOGY AND SOFTWARE DEVELOPMENT

Lead author

itsdm: An R package to use an interpretable Isolation Forest model for species distribution modeling.

hrlcm: A high resolution land cover mapping workflow of using ensemble labels, Random Forest, and U-Net.

sentinelPot: A python package to pre-processing Sentinel-1&2 level-1 images.

waspire: A template to build docker image to run WASP (Weighted Average Synthesis Processor) to create cloud-free syntheses with Sentinel-2 level-2A.

Co-author

labeller: Labelling platform for Mapping Africa active learning project.

learner: Machine learning component of the active learning project.

imager: A repo to prepare PlanetScope images for labelling platform and modeling.

PROFESSIONAL SERVICE

Invited Editorial Positions

2025-Present Associate Editor – *Frontiers in Artificial Intelligence | Specialty: AI in Food, Agriculture and Water*

UNIVERSITY & SCHOOL SERVICE

2019 Promotion Committee, Graduate School of Geography, Clark University

PROFESSIONAL AFFILIATIONS

Global Land Programme
American Association of Geographers (AAG)
American Geophysical Union (AGU)
The Ecological Society of America (ESA)
North American Regional Association of the International Association for Landscape Ecology (IALE-NA)

PEER REVIEWER

Nature Food
Landscape Ecology
Global Change Biology
Remote Sensing of Environment
Environmental Earth Sciences
Remote Sensing
PLOS Climate

Natural Resources Forum
Ecology and Society
PeerJ – Life and Environment
Frontiers in Artificial Intelligence, AI in Food, Agriculture and Water